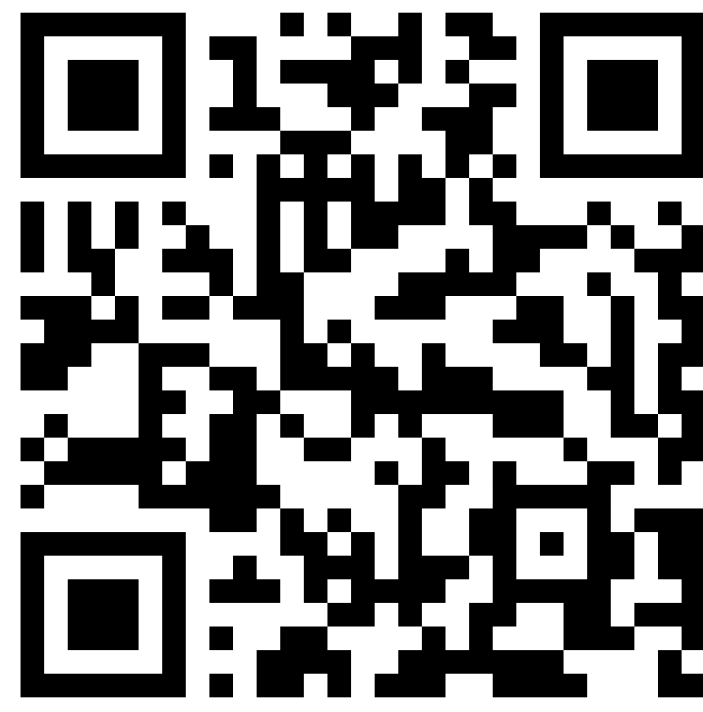




TED University
Department of Computer
Engineering



Our Web Page

Caner Aras
Emir Irkılata
Oğuzhan Özkaya

Group 4

Supervisor:
Ayşenur Birtürk



MoonAI
Neuroevolution Simulation
Platform

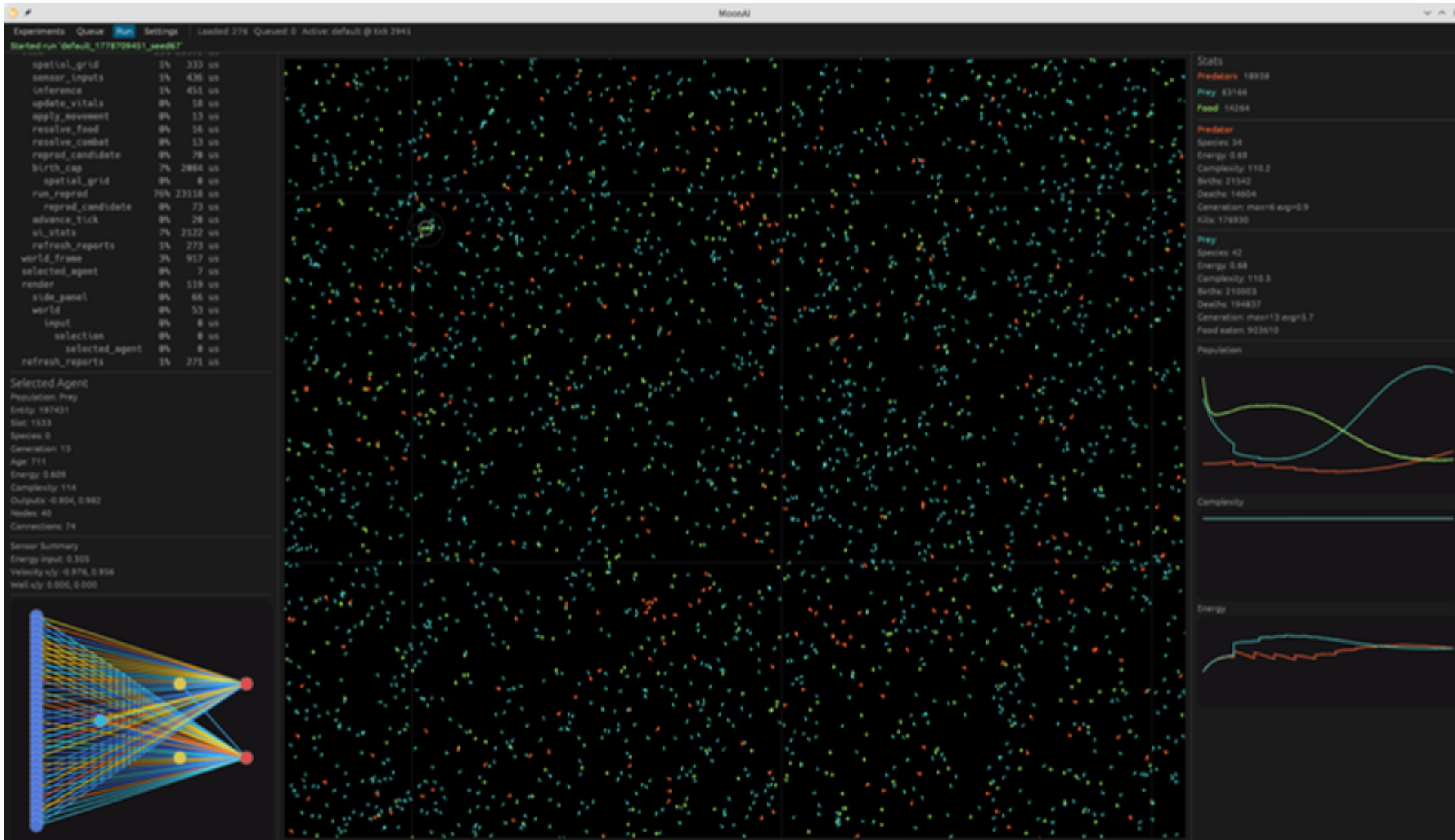
OBJECTIVE

Modern artificial intelligence training requires vast amounts of real-world data, which do not scale efficiently. MoonAI addresses this limitation decoupling training from real-world data dependencies through providing self-generating training environments for studying artificial intelligence without external data dependencies.

PROJECT

MoonAI uses a predator-prey environment as a synthetic benchmark to evaluate evolutionary computation methods.

Agents (predators and prey) are controlled by neural networks whose structure and weights evolve continuously through births and deaths using the NeuroEvolution of Augmenting Topologies (NEAT) algorithm.



Methodology / NEAT + System Architecture

MoonAI simulates a predator-prey environment where agents are controlled by neural networks. Across generations, NEAT improves these networks through mutation, crossover, selection, and fitness-based survival.



Technologies	Results	Contribution
Rust	• Real-time predator-prey simulation was implemented.	MoonAI combines neuroevolution, predator-prey simulation, and GPU acceleration in a single configurable platform. Unlike traditional dataset-based AI systems, MoonAI allows intelligent behavior to emerge naturally through interaction, adaptation, and evolution.
CUDA / GPU Computing	• Agents make decisions using neural-network outputs.	
Wgpu Visualization	• Fitness values are calculated from survival, kills, energy, distance, and complexity.	
NEAT Algorithm	• Experiment data can be exported for later analysis.	
Lua Configuration		
CSV / Data Logging		

FEATURES

Configuration

All of the Simulation parameters are configurable through Lua based configuration.

NEAT

NeuroEvolution of Augmenting Topologies is a genetic algorithm for evolving both the topology and weights of artificial neural networks.

Performance

MoonAI uses CUDA for all of the simulation, evolution, and neural network calculations.

Data Export

The system logs telemetry, including population metrics and genome histories, exporting structured data for external analysis.

Conclusion

MoonAI demonstrates how intelligent behaviors can emerge through evolutionary simulation in a predator-prey ecosystem. By combining NEAT, GPU acceleration, and real-time visualization, the platform provides an efficient environment for studying adaptive AI behavior and neuroevolution.

Future Work

- ➔ More complex environments and maps
- ➔ Additional agent interaction types
- ➔ Improved visualization tools
- ➔ Support for alternative evolutionary algorithms
- ➔ Larger-scale GPU simulations

